

Chapter 9

From the short to the medium run: the *IS-LM-PC* model

Plan

- Develop the IS-LM-PC model
- Study the dynamics of adjustment of output and inflation
- Study the dynamic effects of a fiscal consolidation
- Study the dynamic effects of an increase in the price of oil

From the short to the medium run: the *IS-LM-PC* model

- In Chapters 3 through 6, we looked at equilibrium in the **goods and financial markets**, and how output is determined in the **short run**
- In Chapters 7 and 8, we looked at equilibrium in the **labor market**
- We now put the two parts together and use this **IS-LM-PC** (PC for Phillips curve) model to characterize the behavior of output **both in the short run and the medium run**

Toward the IS-LM-PC model

- From Chapter 6, output in the short run is determined by demand (IS curve):

$$Y = C(Y - T) + I(Y, r + x) + G$$

- From Chapter 8, the relation between inflation and unemployment is given by the Phillips curve:

$$\pi - \pi^e = -\alpha(u - u_n)$$

- We will show that we can write the Phillips curve in terms of output, as follows

$$\pi - \pi^e = (\alpha / L) \underbrace{(Y - Y_n)}_{\text{output gap}}$$

- This way both the IS curve and the Phillips curve are in terms of output

The IS-LM-PC model

- **IS curve**: captures **short-run aggregate demand** forces

$$Y = C(Y - T) + I(Y, r + x) + G$$

- **LM curve**: captures **short-run** effects of **monetary policy**

$$r = \bar{r}$$

- **PC**: captures **medium-run aggregate supply** forces

$$\pi = \pi^e - \alpha(u - u_n)$$

$$\pi = \pi^e + (\alpha/L)(Y - Y_n)$$

The IS-LM-PC model

- Model

$$Y = C(Y - T) + I(Y, r + x) + G$$

$$r = \bar{r}$$

$$\pi = \pi^e + (\alpha/L)(Y - Y_n)$$

- Endogenous variables, in green:
 - Output: Y
 - Interest rate: r
 - Inflation: π
- Policy variables, in red:
 - Fiscal policy: T, G
 - Monetary policy: $\bar{r}$
- The other exogenous variables or parameters, in blue:
 - The labour force: L
 - The risk premium: x
 - The natural level of output: Y_n (in turn function of m and z)
 - The expected inflation rate: π^e
 - The parameter of the PC: α

Derivation of the output gap

- Unemployment rate: $u = \frac{U}{L} = \frac{L - N}{L} = 1 - \frac{N}{L}$

- Rearranging: $N = L(1 - u)$

- Using also $Y = N$: $Y = L(1 - u)$

- This implies that: $Y_n = L(1 - u_n)$

where Y_n = **natural level of output** or **potential output**

- **Output gap:** $Y - Y_n = L(1 - u) - L(1 - u_n) = -L(u - u_n)$

Derivation of the PC in terms of output

- PC in terms of the unemployment gap:

$$\pi - \pi^e = -\alpha(u - u_n)$$

- Substitute $(Y - Y_n) = -L(u - u_n)$:

$$\pi - \pi^e = (\alpha / L) \underbrace{(Y - Y_n)}_{\text{output gap}}$$

- Assume expected inflation this year = inflation last year:

$$\pi - \pi(-1) = (\alpha / L)(Y - Y_n)$$

which says that **when Y is above Y_n (positive output gap), inflation increases**

What does the PC tell us?

- Phillips curve in terms of the output gap:

$$\pi = \pi_{-1} + (\alpha/L)(Y - Y_n)$$

- Implications:

If $Y > Y_n$ then $\pi > \pi(-1)$, that is, inflation is increasing

When output is above potential the economy is in an expansion, the economy is overheated, and there are upward pressures on wages and prices

If $Y < Y_n$ then $\pi < \pi(-1)$, that is, inflation is decreasing

When output is below potential the economy is in a recession, the economy is depressed, and there are downward pressures on wages and prices

If $Y = Y_n$ then $\pi = \pi(-1)$, that is, inflation is constant

When output is at potential the economy is at the “natural” equilibrium and inflation is constant

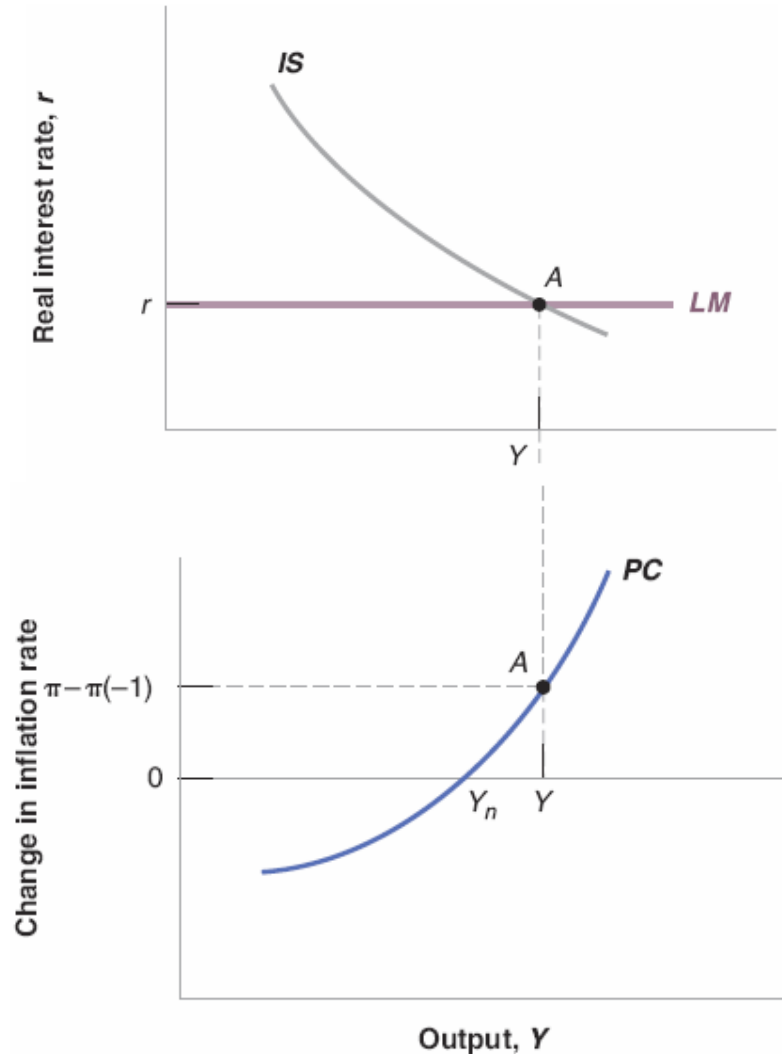
The IS-LM-PC model

$$Y = C(Y - T) + I(Y, r + x) + G$$

A lower policy rate leads to higher output (lower r , higher I , higher Z , and higher Y)

$$\pi = \pi_{-1} + (\alpha/L)(Y - Y_n)$$

- A higher output leads to a larger change in inflation
- The change in inflation is 0 when the output gap is 0, that is, when output equals potential output
- The change in inflation is positive (negative) when output is above (below) potential



The IS-LM-PC model

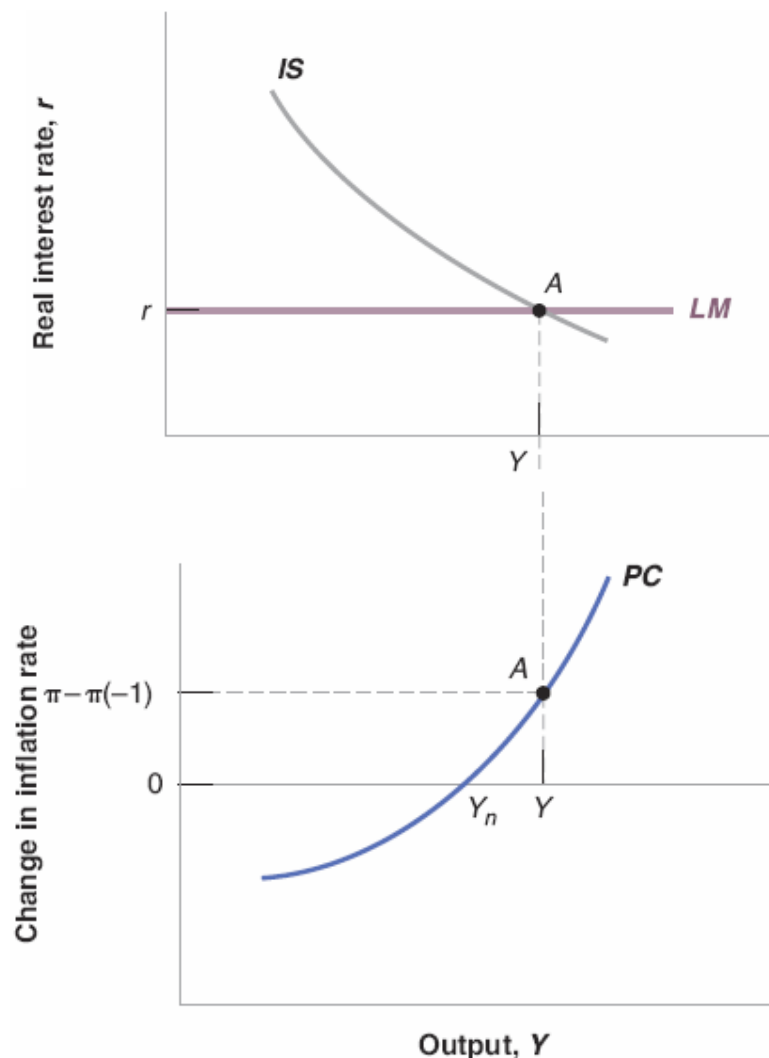
Equilibrium A is a short run equilibrium

$$Y = C(Y - T) + I(Y, r + x) + G$$

- Suppose the central bank picks a policy rate equal to r
- Associated with this interest rate there is a level of output Y

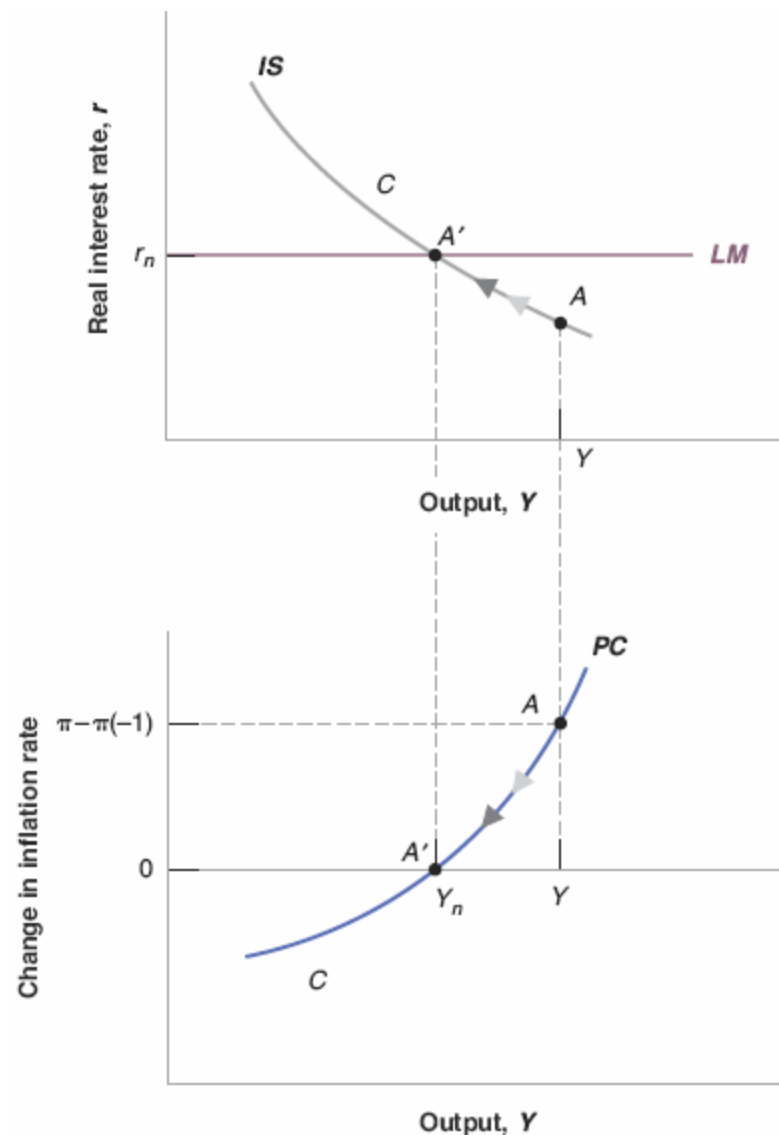
$$\pi = \pi_{-1} + (\alpha/L)(Y - Y_n)$$

- The level of output Y is above potential and implies that inflation is increasing
- Less formally: the economy is overheated, putting pressures on inflation



The IS-LM-PC model: dynamics and medium run equilibrium

- If there is no change in the monetary policy rate, nor in fiscal policy, output remains above potential and inflation continues increasing
- At some point, **the central bank (CB) will respond**: it will **gradually increase the policy rate** to **decrease output to potential** output and to **eliminate inflationary pressures**
- The initial equilibrium is in point A. As the CB increases the policy rate r , the economy moves from equilibrium A to equilibrium A'
- From the top graph we see the economy moves along the IS curve and output decreases
- From the bottom graph we see that as output decreases, the economy moves along the PC and the change in inflation becomes smaller (despite positive)
- **At point A', the medium run equilibrium, output equals potential and the change in inflation is 0 (inflation is constant)**
- The policy rate in the medium run equilibrium, associated to potential output Y_n , is denoted r_n and is called **the natural rate of interest** (or the neutral or Wicksellian rate of interest)



The IS-LM-PC model: dynamics and medium run equilibrium

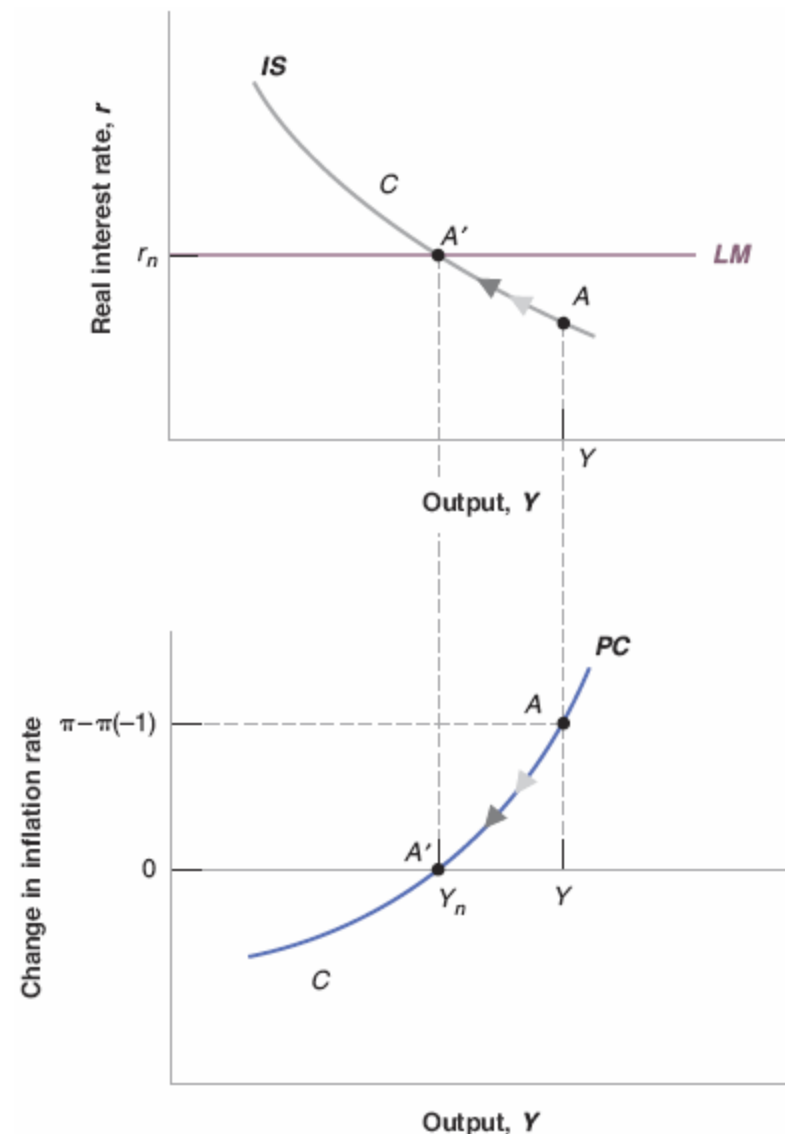
- Two key observations
- **First**, if the central bank wants to keep output at potential and inflation stable, why doesn't it do it right away? Because this is not as easy as it seems:
 - First, Y_n is not observed and it is not always easy to estimate
 - Second, it takes time for the economy to respond
- **Second**, note that while the central bank raises the policy rate gradually and output decreases to Y_n , **inflation is continuously increasing**

Thus, **when the economy reaches equilibrium A' , inflation is higher than in A !**

If the central bank cares not only about **stable inflation**, but also about the **level of inflation**, it will have to reduce inflation

To do this, it needs to increase the policy rate beyond r_n to generate a decrease in inflation **until the level of inflation is back to an acceptable level**, reaching for example point C

Put it differently, if the central bank wants to achieve a constant level of inflation over the medium run, then the initial boom must be followed by a recession



Short-run equilibrium in the IS-LM-PC model

$$\pi \neq \pi^e \text{ (or } P \neq P^e)$$

$$Y \neq Y_n \text{ (or } u \neq u_n)$$

$$r \neq r_n$$

Price rigidity:

- In the “IS-LM” model **prices** are completely **fixed**
- In the “IS-LM-PC” model, prices exhibit partial rigidity—they are **sticky**. Specifically, wages do not adjust flexibly to unexpected shocks, meaning **wages** are **sticky**. As a result, the expected price P^e used in wage-setting may differ from the actual price P

Role of demand and of monetary and fiscal policies:

- When prices are rigid, demand factors influence output in the short run
- Among these factors, monetary and fiscal policies play a significant role
- If **monetary policy** impacts output, it is said to have **real effects**, or that it is **non-neutral**

Medium-run equilibrium in the IS-LM-PC model

$$\pi = \pi^e \text{ (or } P = P^e \text{)}$$

$$Y = Y_n \text{ (or } u = u_n \text{)}$$

$$r = r_n$$

Price flexibility:

- In the medium run, **prices** adjust and become flexible **flexible**
- Agents, particularly wage setters, adjust their expectations to align with actual prices
- In the "IS-LM-PC" model, wage setters renegotiate wages in the medium run, ensuring that the expected price P^e used in wage setting aligns with the actual price P

No role of demand and monetary neutrality:

- When prices are flexible, demand factors do not influence output
- Output equals the natural level of output, Y_n , determined by non-demand factors such as labor market institutions, z , and degree of competition in goods markets, m
- The "WS-PS" model describes this medium run equilibrium, determining u_n (and Y_n)
- If **monetary policy** cannot affect output, it is said to be **neutral**

From the short-run to the medium-run equilibrium in the IS-LM-PC model

There are two ways the economy can transition from short-run to medium-run equilibrium:

Adjustment of price expectations:

- Agents, particularly wage setters, revise their expectations to align with actual prices
- We do not study this case

Monetary policy stabilization:

- The central bank aims to maintain stable inflation and ensure output remains at its natural level; to do that, it adjusts the policy rate accordingly
- Fiscal policy can also contribute to stabilization, but it is less flexible as a tool

In this course, the focus is on **monetary policy stabilization**, where **the interest rate (r) is adjusted to its natural level (r_n)**, as it is considered the most realistic approach

Revisiting the role of expectations in the PC

- Recall: before 1970s inflation expectations set at the average level of inflation

$$\pi_t^e = \bar{\pi}$$

- After 1970s, wage setters changed the way of setting expectations

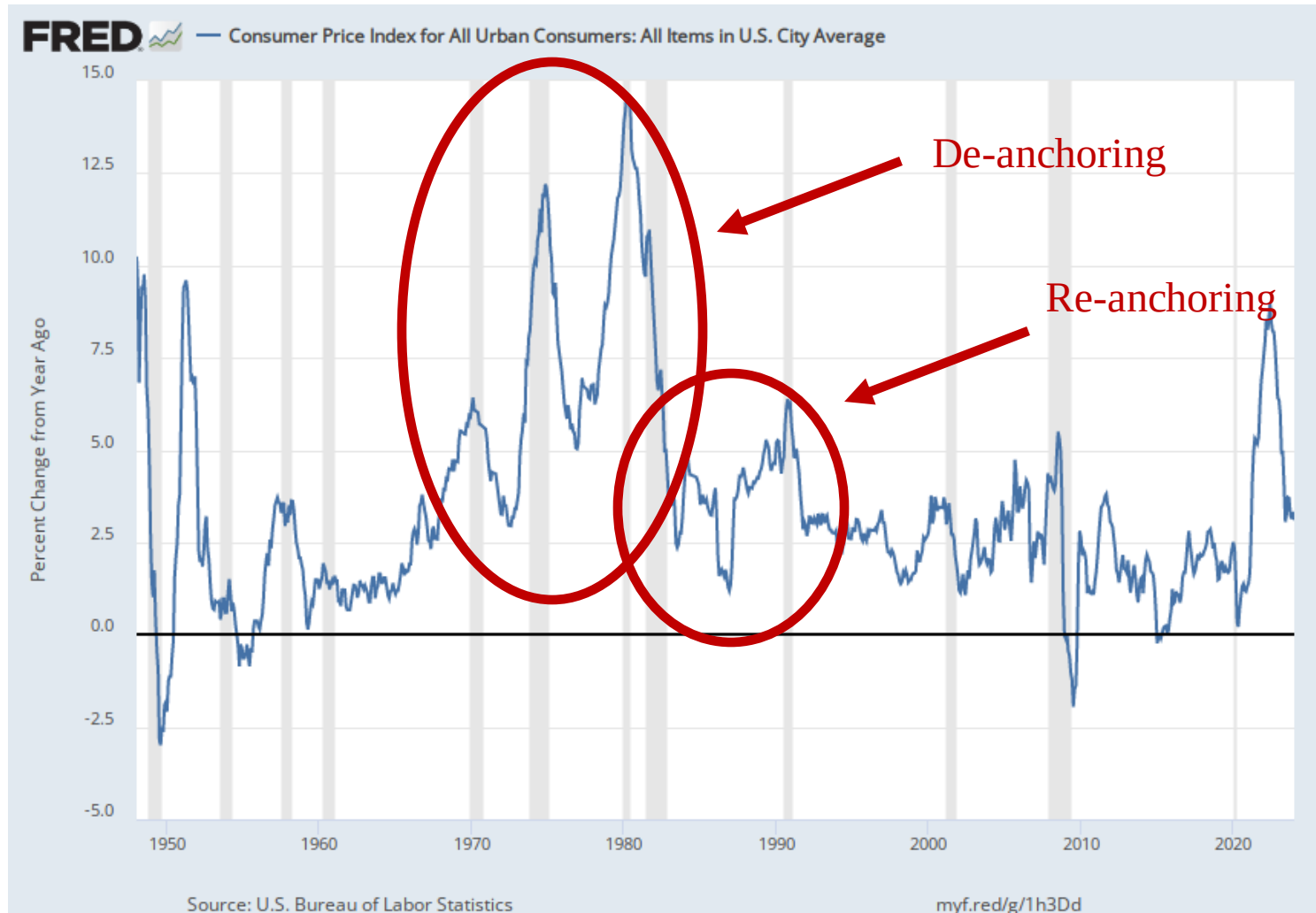
$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- In macro jargon, we refer to this phenomena as **de-anchoring of expectations**
- Expectations previously anchored at $\bar{\pi}$, de-anchor and starts being determined by current inflation, so that higher inflation leads to higher expected inflation
- In the 1990s, inflation expectations **re-anchored** to the **target inflation rate**

Re-anchoring of inflation expectations in the 1900s

- Starting the early 1980s, many CBs (including the Fed) increasingly emphasized their **commitment to maintaining low and stable inflation**
- Many indicated that they would maintain inflation close to a **target value, typically 2%**
- By the 1990s, the Fed had largely achieved this goal, and inflation had been stable for a decade so that the **target** became **credible**
- Wage setters **changed again the way they formed expectations**
- Even if inflation was higher than the target for a year, people assume that the CB would take measure to bring inflation back to target in the future
- Expected inflation become roughly constant, and equal to the target set by the CB
- **Expectations** that had become de-anchored in the 1970s and 1980s, **become re-anchored** from the 1990s, hence reacting little or not at all to movements in actual inflation

Inflation rate, US, 1948-2024



Implications of anchored expectations

- Recall that if inflation expectations are anchored at some $\bar{\pi}$, the PC is

$$\pi = \bar{\pi} + (\alpha/L)(Y - Y_n)$$

- Implications:

If $Y > Y_n$ then $\pi > \bar{\pi}$, that is, inflation is higher than target inflation

When output is above potential the economy is in an expansion, the economy is overheated, and there are upward pressures on wages and prices

If $Y < Y_n$ then $\pi < \bar{\pi}$, that is, inflation is lower than target inflation

When output is below potential the economy is in an expansion, the economy is depressed, and there are downward pressures on wages and prices

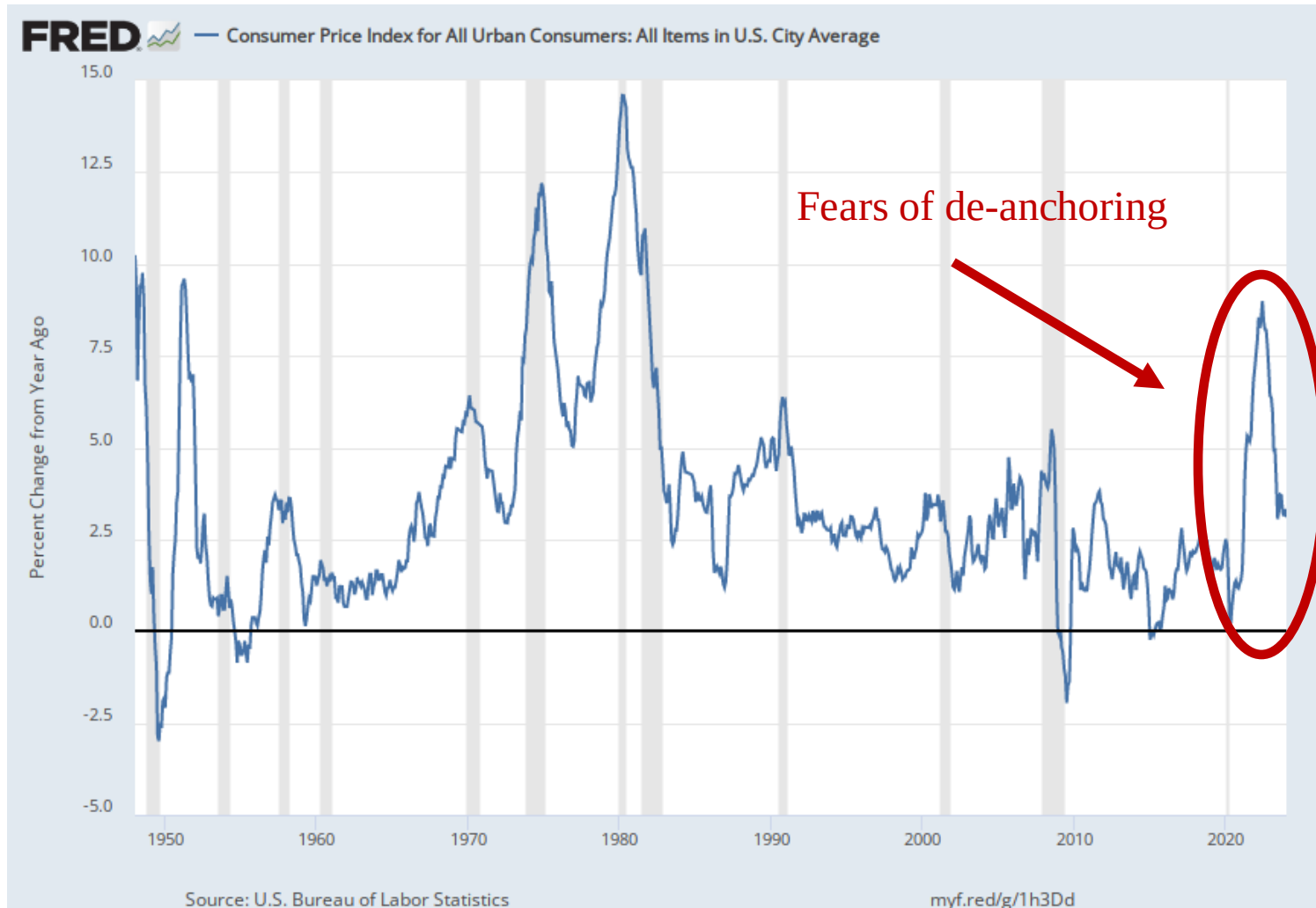
If $Y = Y_n$ then $\pi = \bar{\pi}$, that is, inflation is equal to target inflation

When output is at potential the economy is at the “natural” equilibrium and inflation is constant

Anchored expectations: implications for inflation dynamics

- As we said, a positive output gap generates **a higher level of inflation**, rather than **an increase in inflation**
- In previous analysis, when the economy reaches the new equilibrium A' , and the policy rate is at r_n , then output is back at potential and inflation is at $\bar{\pi}$
- So long as inflation **expectations** remain **anchored**, the central bank does not need to compensate for the initial boom by a recession later
- **Credible** CBs have an easier time, in normal times, to maintain **expected inflation anchored at** the **target level for inflation**, typically 2%

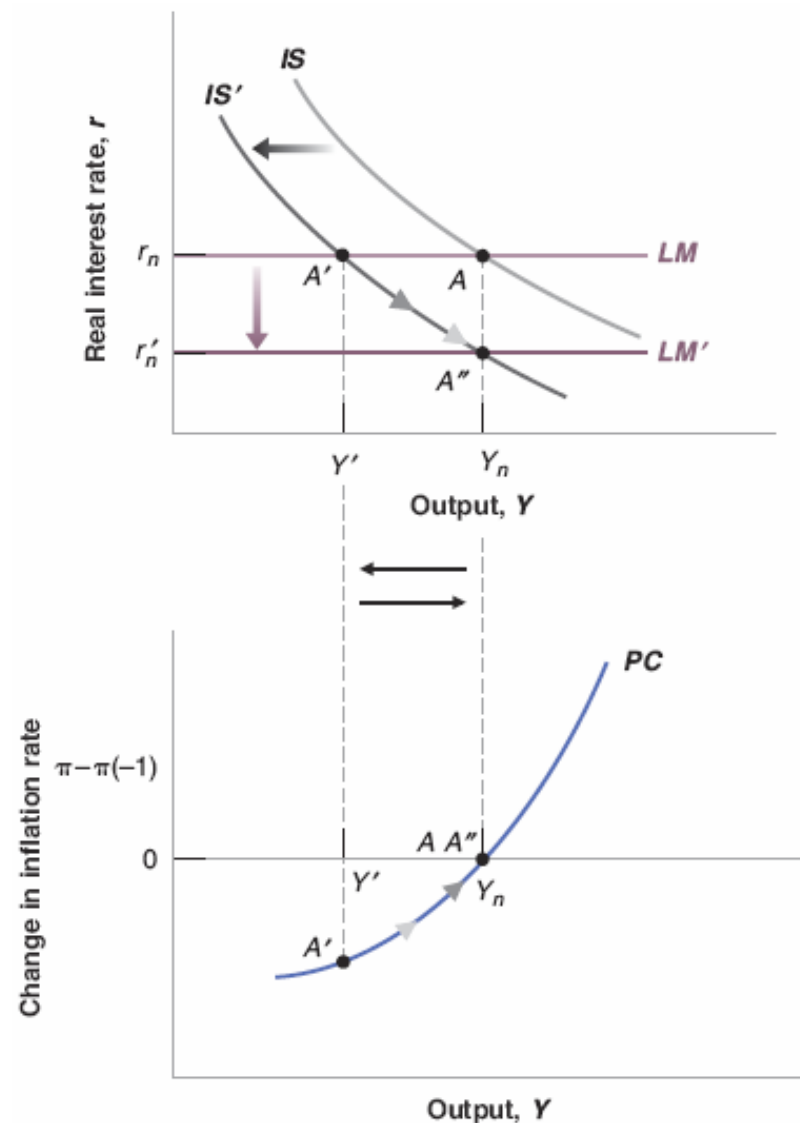
Inflation rate, US, 1948-2024



Recent inflation surge associated to fears of de-anchoring of expectations!

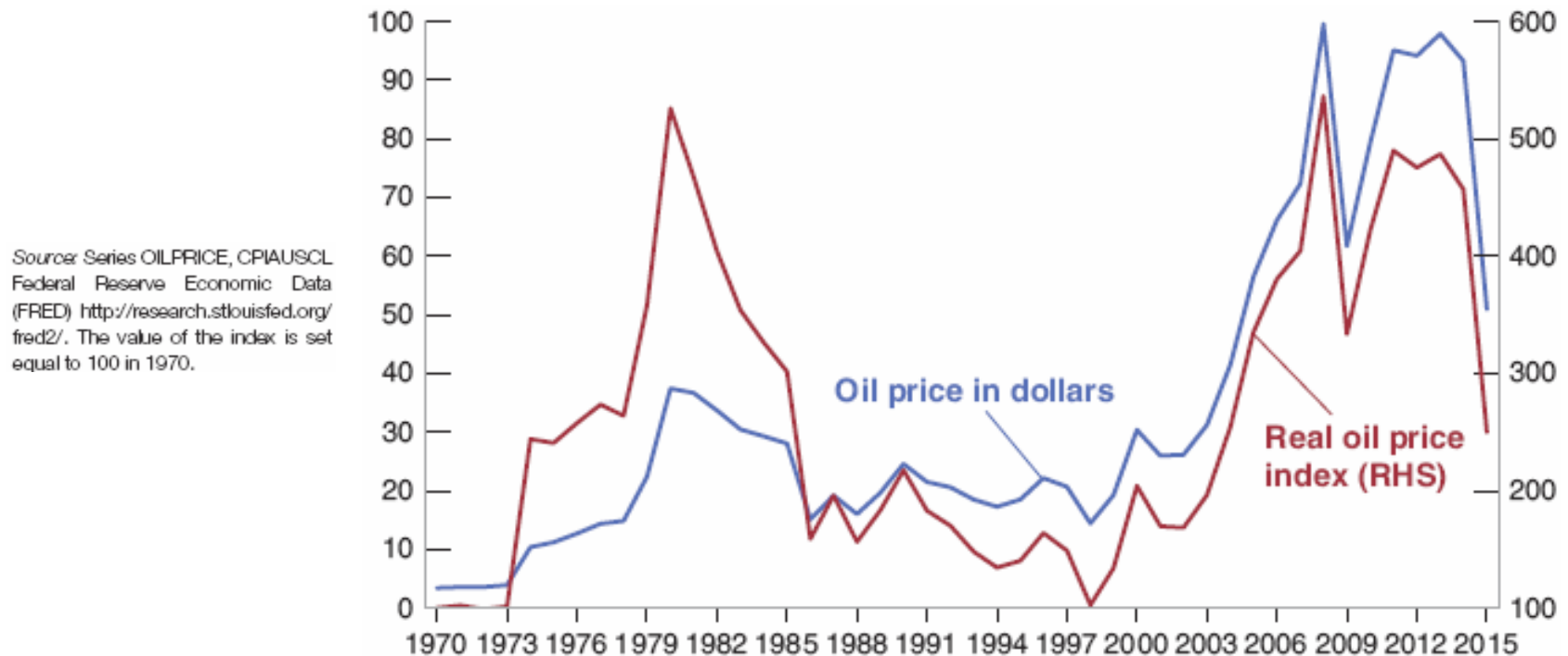
A fiscal consolidation in the short and in the medium run

- Suppose output is at potential and the economy is in point A: $Y=Y_n$, $r=r_n$ and inflation is stable
- Now the government implements a **fiscal consolidation**: it **increases taxes T**
- The IS curve shifts to the left. The new equilibrium is in A'. At the initial policy rate r_n , output decreases (top graph) and inflation starts decreasing (bottom graph). The fiscal consolidation leads to a recession. A' is the short run equilibrium studied in Chapter 5
- The central bank is however likely to react to bring output back at potential. The LM shifts down and the economy moves to equilibrium A''. Output increases back to potential (top graph) and inflation is again stable (bottom graph)
- After the fiscal consolidation, the interest rate needed to maintain output at potential is lower, r'_n rather than r_n
- What do we know about **composition of output**?
 - Short run**: C lower because of higher T and lower Y ; I lower because of lower Y
 - Medium run**: C lower because of higher T ; I higher because of lower r
 - The fiscal consolidation reduces investment in the short run, but increases investment in the medium run if the central bank can coordinate with the fiscal authorities!



The effects of an increase in the price of oil, a supply shock

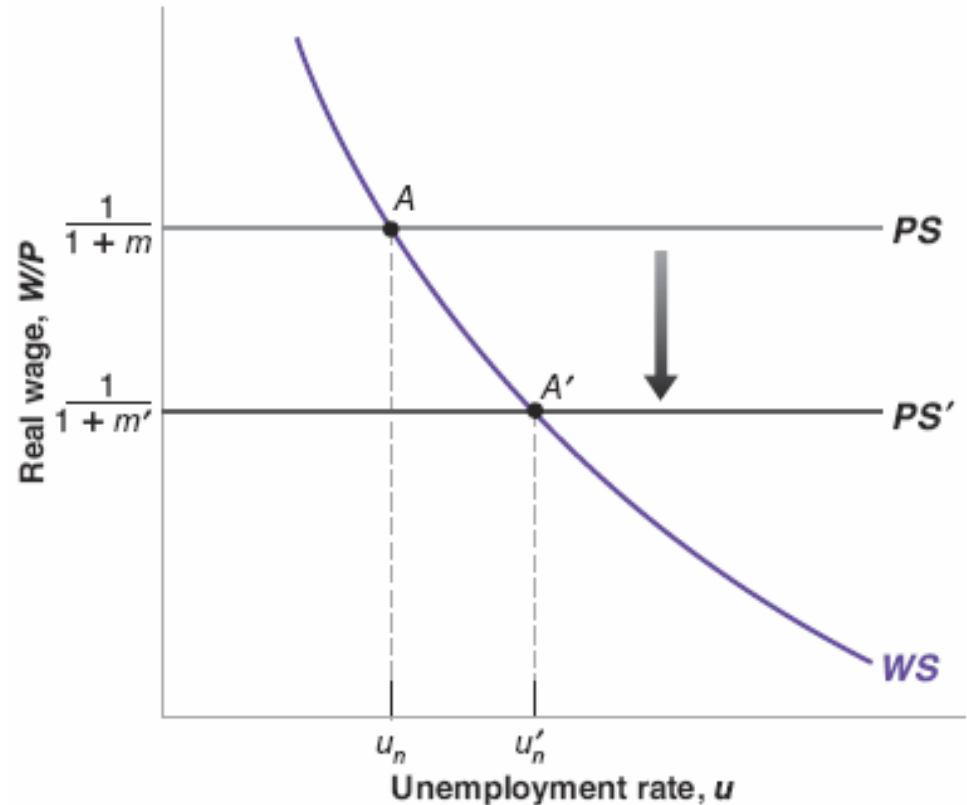
The nominal and the real price of oil, 1970–2015



Over the last 50 years, there have been two sharp increases in the real price of oil, the first in the 1970s (formation of OPEC) and the second in the 2000s (higher demand from emerging economies, in particular China)

The effects of an increase in the price of oil on the natural rate of unemployment

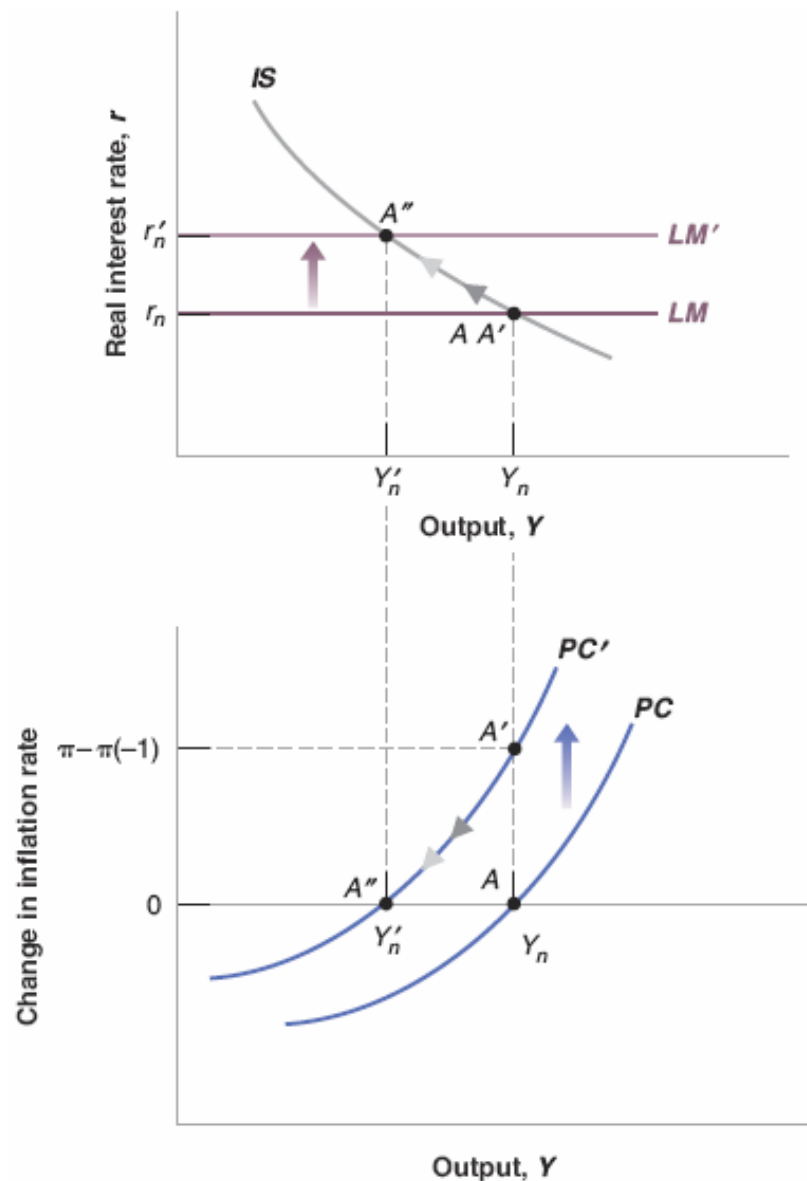
- Where is the price of oil in our model?
- **Oil** is an **input to production**, but we have assumed Y is produced using only N
- Can we study the effects of a change in the price of oil without changing the model? What happens when the **price of oil raises**?
- Even if wages don't change, it is more expensive to produce, that is, the **cost of producing an additional unit of Y increases**
- Thus, firms will **charge a higher price** at given wages...
- This has the **same effect as an increase in the markup!**



- An **increase in the price of oil** is equivalent to an **increase in the markup**
- It leads to lower real wages and a higher natural rate of unemployment
- **Supply shocks change the natural rate of unemployment u_n and potential output Y_n**

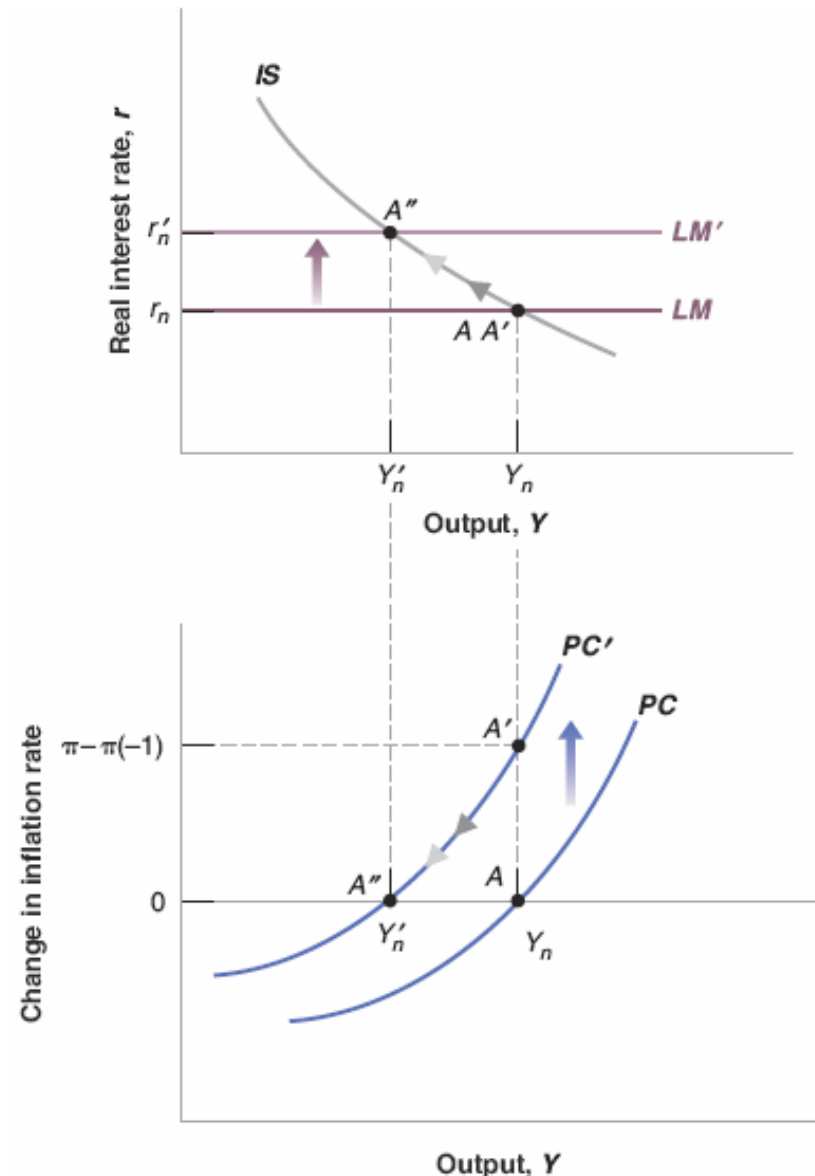
Short and medium run effects of a raise in the price of oil

- Suppose output is at potential and the economy is in point A: $Y=Y_n$, $r=r_n$ and inflation is stable
- Now the **price of oil increases**: the natural rate of unemployment goes up and **potential output goes down**
- **The PC curve shifts up**. The new equilibrium is in A'. At the initial policy rate r_n and if the **IS curve does not shift**, output is unchanged, but the associated inflation is higher. For given wages, the higher price of oil leads firms to increase their prices, so inflation is higher
- If the central bank were to leave the policy rate unchanged, output would remain higher than the now lower potential output and inflation would continue increasing
- The central bank is however likely to react to stabilize inflation. The LM shifts up and the economy moves to equilibrium A''. Output decreases to the new potential level and inflation increases but more and more slowly until it is again stable
- **Stagflation** - lower output and higher inflation - occurs along the transition



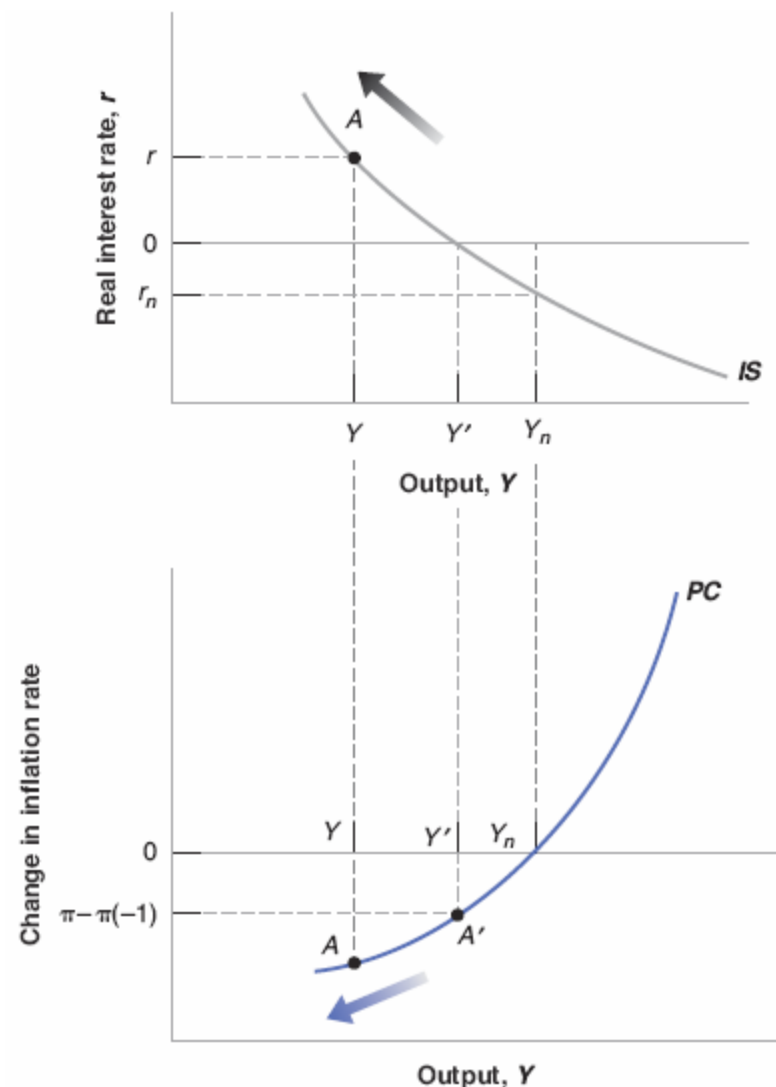
Short and medium run effects of a raise in the price of oil

- Three key observations
- **First**, when the price of oil increases there are several reasons demand could also change, shifting the IS curve
 - First, firms could reduce their investment plans
 - Second, the increase in the price of oil redistributes resources from oil producers to oil buyers. If oil producers consume less than oil buyers, overall demand will go down
 - If, for the reasons above, the IS curve shifts to the left, then **output goes down also in the short run**
- **Second**, consider the evolution of inflation
 - In A'' inflation will be higher than in A , before the raise in the price of oil, since inflation has been continuously raising
 - To lower inflation back to the initial level, the central bank must reduce Y below the new natural level for some time
 - In this case, the economy will go through a **larger recession** with **only a partial recovery**
- **Third**, consider the role of price expectations
 - Suppose expected inflation is constant, as opposed to equal to last year inflation (because inflation targeting and CB credibility)
 - Then when the economy reaches A'' , inflation is back to the level before the increase in the price of oil
 - There is no need for the central bank to further decrease output to decrease inflation



The zero lower bounds and deflation spiral

- Consider the case in which the economy is in a recession: at equilibrium A, at the current policy rate r , output is far below Y_n
- What should the central bank do? Reduce the policy rate until output is back to potential, that is, until $r=r_n$
- If the **zero lower bound** prevents monetary policy from reducing r to r_n and increasing output back to potential, the result may be a **deflation spiral** or a **deflation trap**
- Assume that initially the inflation rate is 0, then because of the zero lower bound the minimum real rate that the central bank can achieve is 0 and an associated level of output Y' still below potential output (equilibrium A')
- Since output is below potential, inflation decreases but as inflation decreases and becomes negative (a deflation) causing the real interest rate to increase and output to decrease further
- More deflation leads to a higher real policy rate, and the higher policy rate in turn leads to lower output and more deflation
- Instead of converging to the medium run equilibrium the economy moves away from it, with output steadily decreasing and deflation steadily becoming larger



Deflation in the Great Depression

- The economy seemed to be in a deflation trap between 1929 and 1933
- Monetary policy decreased the nominal interest rate from 5.3% to 2.6% in 1933, but with negative inflation rates, the real rate reached 12.3% in 1931 and 7.8% in 1933
- Deflation turned larger and larger, output growth turned increasingly negative and unemployment increasingly high

The Nominal Interest Rate, Inflation, and the Real Interest Rate, 1929–1933

Year	Unemployment Rate (%)	Output Growth Rate (%)	One-Year Nominal Interest Rate (%), i	Inflation Rate (%), π	One-Year Real Interest Rate (%), r
1929	3.2	−9.8	5.3	0.0	5.3
1930	8.7	−7.6	4.4	−2.5	6.9
1931	15.9	−14.7	3.1	−9.2	12.3
1932	23.6	−1.8	4.0	−10.8	14.8
1933	24.9	9.1	2.6	−5.2	7.8

Deflation in the Great Recession

- The financial crisis gave rise to similar worries
- With the nominal policy rate at the zero lower bound in major advanced economies, the worry was that inflation would turn negative and start a similar spiral
- This did not happen. Inflation decreased and, in some countries, it turned negative, but **deflation was limited**, and the deflation spiral did not happen. Why?
- One reason is that inflation **expectations remained largely anchored**
- As a result, the Philips curve equation took the form of

$$\pi = \bar{\pi} + (\alpha/L)(Y - Y_n)$$

- Negative output gaps led to low inflation and, in some cases, to mild deflation, but not to a deflation trap as it was the case during the Great Depression
- **Credibility** of modern central banks has been key

The IS-LM-PC model with math

- **IS curve**: captures **short-run aggregate demand** forces

$$Y = C(Y - T) + I(Y, r + x) + G$$

- **LM curve**: captures **short-run** effects of **monetary policy**

$$r = \bar{r}$$

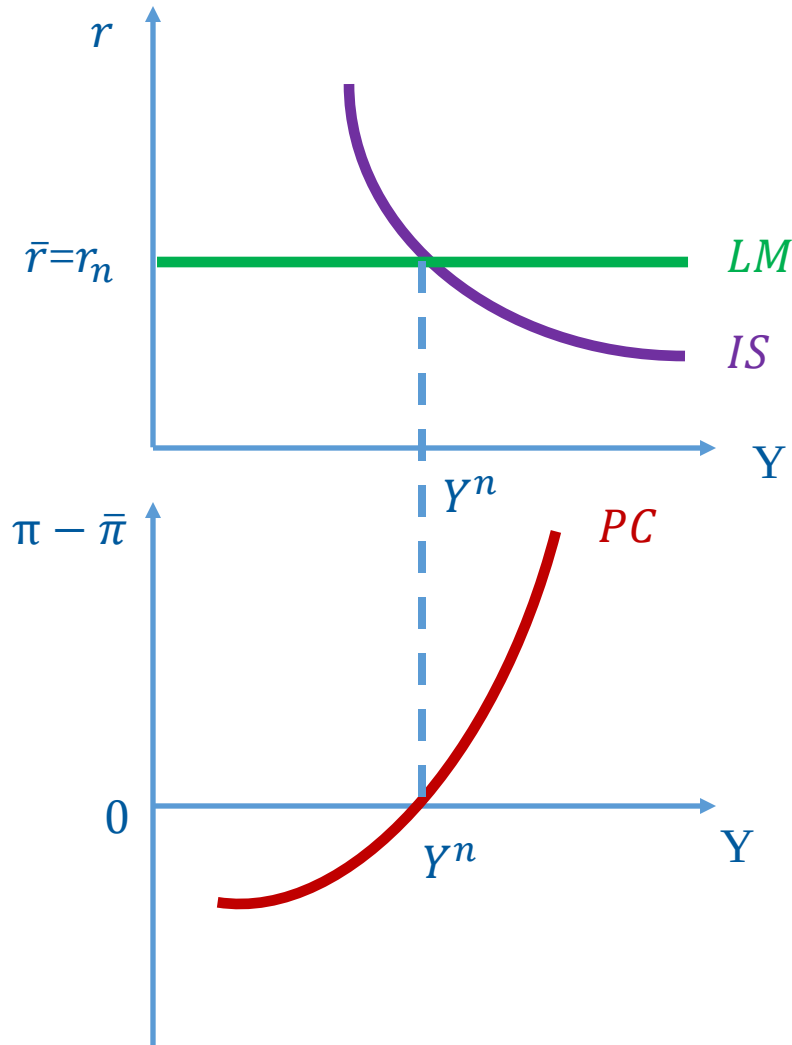
- **PC**: captures **medium-run aggregate supply** forces

$$\pi = \pi^e + (\alpha/L)(Y - Y_n)$$

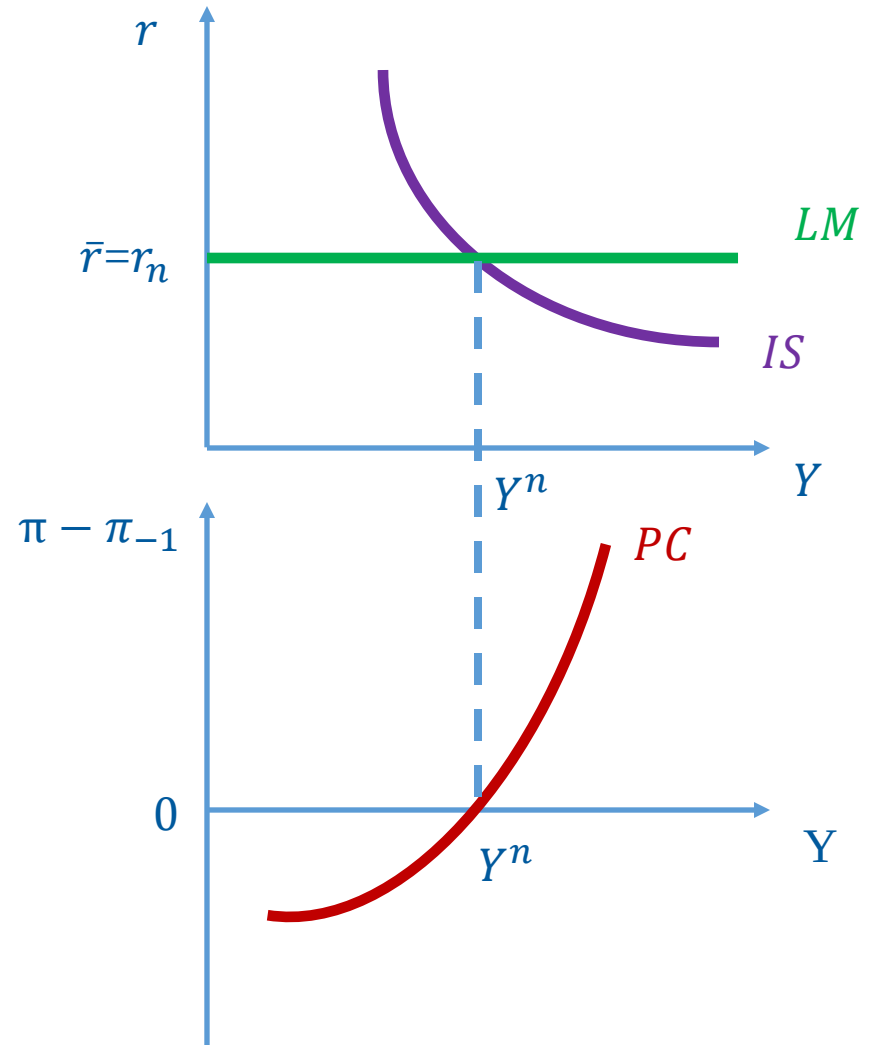
- If $\pi^e = \bar{\pi} \rightarrow \pi = \bar{\pi} + (\alpha/L)(Y - Y_n) \rightarrow$ PC is a relation between $\pi - \bar{\pi}$ and $Y - Y_n$
- If $\pi^e = \pi_{-1} \rightarrow \pi = \pi_{-1} + (\alpha/L)(Y - Y_n) \rightarrow$ PC is a relation between $\pi - \pi_{-1}$ and $Y - Y_n$

The IS-LM-PC model with graphs

Medium run equilibrium



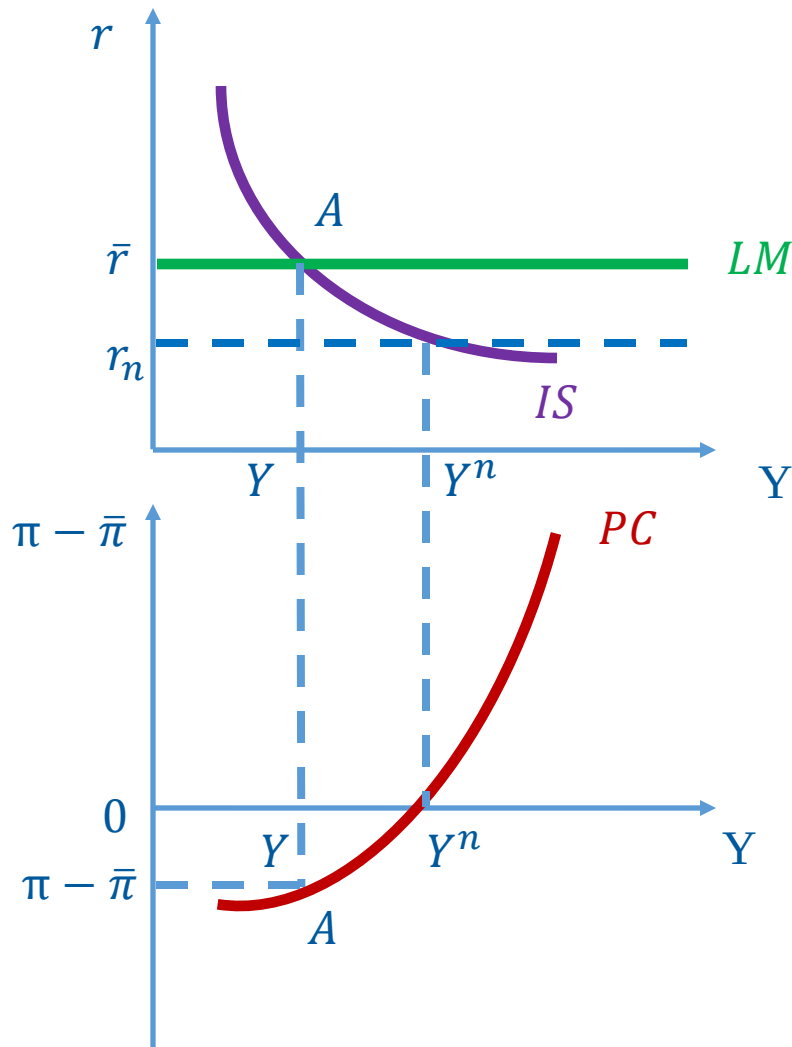
PC with **anchored** expectations



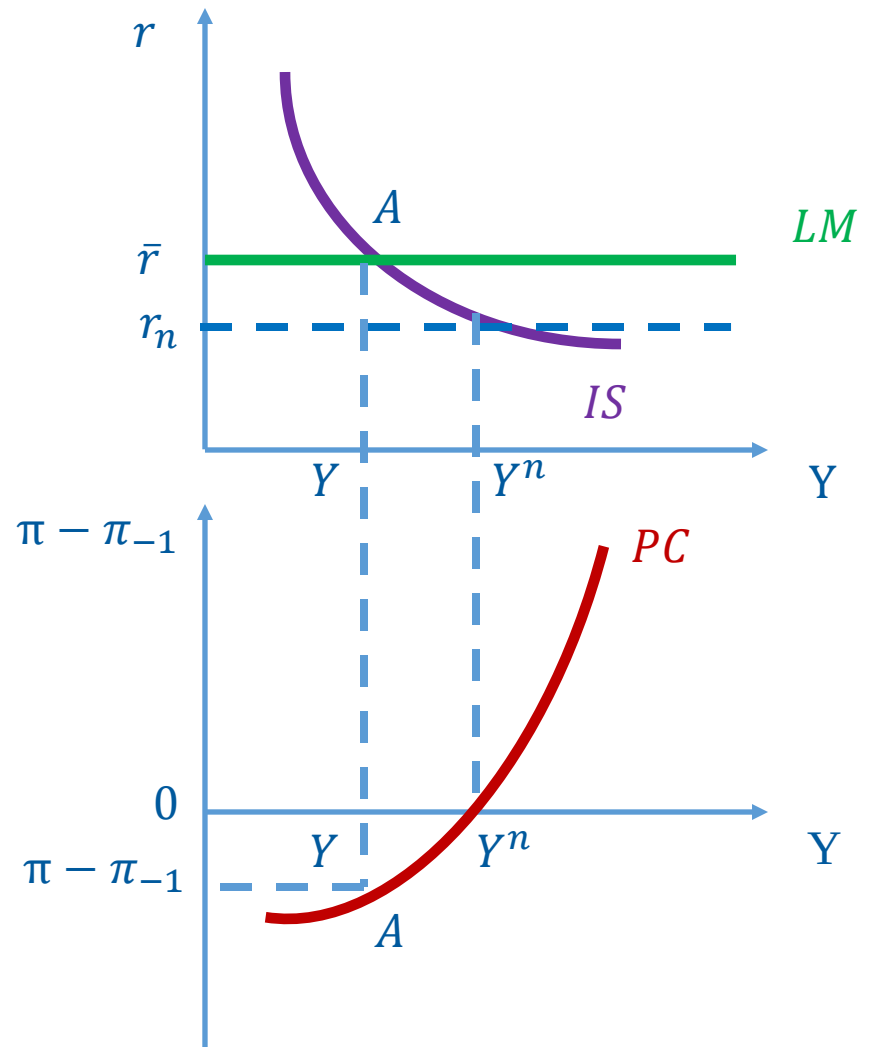
PC with **unanchored** expectations

The IS-LM-PC model with graphs

Short run equilibrium: recession



PC with anchored expectations



PC with unanchored expectations

Summing up and conclusions

- Shocks or changes in policy typically have different effects in the short run and in the medium run
- Disagreements about the effects of various policies depend on how fast the economy is thought to adjust to shocks, that is, whether the focus is on the short or the medium run
- Movements in output around its trend are called **output fluctuations** or **business cycles**
- **Economic fluctuations** are the results of shocks and their dynamic effects, called the **propagation mechanism**

Summing up and conclusions

- We have thought of **monetary policy** as the key policy tool to bring the economy back to its **medium run equilibrium**
- However, **fiscal policy** can also be used as a tool to **stabilize business cycles**
- We have looked at **two** types of **shocks** causing short run changes in output and inflation: a **fiscal consolidation** and an **increase in the price of oil**
 - We have assumed monetary policy moves to offset the short run effects
 - A key difference between the two types of shocks is that the first only changes aggregate demand, while the second also changes the natural level of output
- **Other types of shocks** are: a **financial shock** (change in x); a **consumer confidence shock** (change in c_0); a change in **labor market institutions** (change in z), etc...
- We studied the consequences of the **ZLB** and the possibility of a **deflation spiral**
- We have looked at **two PC** associated to two ways of setting **inflation expectations**